Hemanjali Reddy

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EXPERIENCE

Pfizer Inc. **New York, NY**
AI Engineer *Jun 2026 – Aug 2026*

- Architected a multi-dimensional **evaluation framework** for an oncology AI assistant integrating LLM-as-a-Judge, trajectory analysis, and behavioral-drift detection to assess response quality and agent execution consistency.
- Developed an LLM-as-a-Judge pipeline with context-aware batch evaluation and tool evidence, benchmarking stochastic judge scores against deterministic metrics to assess evaluator reliability.
- Designed **perturbation-based** robustness tests to quantify model sensitivity to input variations and surface paraphrase-sensitive failure modes.
- Built **trajectory analysis** pipelines using vector representations, UMAP, clustering, and geometric heuristics to characterize agent execution patterns and analyze behavioral consistency across runs.

ResMed Technologies **Bengaluru, India**
Associate Machine Learning Engineer *Jul 2024 – Jul 2025*

- **Built the agent harness** for a production **agentic medical chatbot infrastructure** using LangChain and LangGraph, exposed via FastAPI microservices on AWS/Docker.The application has since served 15.9M+ requests to date.
- Engineered the chatbot's context layer and **agent scaffolding**, integrating customer-specific instructions, clinical context, retrieval, and safety constraints for grounded, context-aware responses.
- Developed an **automated regression-evaluation system** to benchmark prompt, model, context, and agent-harness changes across quality, latency, safety, and failure modes, saving experiment runtime and guiding prompt selection for **improved customer-facing responses.**
- Led migration from Guardrails AI to AWS Bedrock Guardrails, **reducing end-to-end latency by 7** seconds while strengthening privacy, safety, and HIPAA-aligned controls.
- Built an agentic **RAG system for insurance claims** that routed queries between a **SQL agent** over claims data and PostgreSQL vector retrieval over HME/DME documentation, followed by LLM synthesis.
- Benchmarked models across **20M+ insurance records** and selected/optimized CatBoost via cross-validation, hyperparameter tuning, and drift analysis to improve classification performance.

Mobileum Technologies **Bengaluru, India**
Machine Learning Intern *Jan 2023 – Jun 2023*

- Trained and benchmarked Decision Tree, Random Forest, and deep learning model for spam-call detection using telecom Call Detail Records, comparing classification performance across approaches.

RESEARCH & TEACHING EXPERIENCE

Cornell University **Ithaca, NY**
Graduate Research Assistant — Comparative Constitutional Analysis with LLMs *Feb 2026 – May 2026*

- Benchmarked LLM performance across **460 variables** and **581 constitutions** against human experts, achieving **71% agreement**; built automated pipelines for legal document ingestion and inter-rater reliability computation.

Cornell University **Ithaca, NY**
Graduate Research Assistant — Multimodal AI for Harmful Speech Detection *Jan 2026 – May 2026*

- Benchmarked vision-language models on real-world, code-mixed multimedia to evaluate robustness across culturally nuanced and contextually implicit harmful-speech categories.

Cornell University **Ithaca, NY**
Teaching Assistant — STSCI 2100: Introductory Statistics and Data Science *Fall 2025*

- Led weekly undergraduate labs in R programming, statistics, and data analysis with hands-on experiments.

PROJECTS

Probabilistic Calibration of LLMs – Los Alamos National Laboratory Capstone **2025 – 2026**

- Built an LLM calibration framework comparing six prompting strategies using Brier score, ECE, and cross-entropy; found calibration error was **6–8x lower** than foundational ML baselines, with self-critique prompting recovering up to **52%** of error in the hardest setting.

InferBench — LLM Inference Performance Lab **2026**

- Built a modular CPU-based LLM benchmarking framework analyzing KV caching and CPU threading, measuring 3.4× faster decoding and 56% lower latency in tested configurations.

EDUCATION

Cornell University **Ithaca, NY**
Master of Professional Studies in Applied Statistics *CGPA: 3.9/4.0 — Aug 2025 – May 2026*

- **Relevant Coursework:** Linear Models with Matrices, Multivariate Analysis, Probability Models, Data Mining, SAS High-Performance Computing

Vellore Institute of Technology **Vellore, India**
B.Tech. & M.Tech. (Integrated), Computer Science Engineering, Minor in Data Science *CGPA: 9.18/10.0 — Sep 2020 – May 2025*

SKILLS & CERTIFICATIONS

Languages & Databases: Python, R, C++, SQL, SAS, PostgreSQL, DynamoDB, MongoDB
AI/ML Frameworks: scikit-learn, PyTorch, TensorFlow, NumPy, Hugging Face Transformers, LangChain, LangGraph, Google ADK
Frontend, Backend & APIs: React.js, HTML/CSS, FastAPI, REST APIs, Streamlit, Linux
MLOps, Cloud & DevOps: AWS, GCP, Docker, GitHub Actions, CI/CD, Datadog, Terraform
Certifications & Awards: AWS Solutions Architect Associate | Machine Learning – Stanford | Data Science – Udemy | **3rd Place, ResMed Hackathon**